

# Divyang Deepak Palshetkar

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## Summary

Fire scientist and environmental engineer (**Dept. Rank 1**, IIT Jodhpur) with hands-on expertise in **coupled fire-atmosphere-chemistry modelling** using **WRF-Chem-Fire / WRF-SFIRE**, wildfire spread dynamics, **aerosol radiative feedback** analysis, and fire-weather interactions in complex terrain. Proficient in **Python, R, GIS** (ArcGIS/QGIS), **NetCDF/GRIB** spatial data pipelines, and statistical/ML-based environmental analysis. Demonstrated ability to independently design, execute, and communicate high-resolution numerical fire simulations — directly aligned with Technosylva's operational wildfire risk and fire behaviour modelling mission.

## Education

**B.Tech-M.Tech Dual Degree, Civil & Infrastructure (Environmental) Engineering** 2021 – 2026 (Expected)  
*Indian Institute of Technology (IIT) Jodhpur* | *B.Tech CGPA: 8.3 / 10* | *M.Tech CGPA: 9.5 / 10* | **Department Rank: 1**

– **Relevant Coursework:** Air Pollution & Control (A), Climate Change & Impact (A), Geo-informatics / GIS (A), AI/ML in Infrastructure Engineering (A), Water Resources Engineering, Environmental Engineering (A), Fluid Mechanics (A-)

## Core Skills

**Fire & Atmospheric Modelling:** WRF-Chem-Fire / WRF-SFIRE (Rothermel fire spread, two-way fire-atmosphere coupling, level-set perimeter tracking), Grid-FDDA data assimilation / nudging, MOZCART photochemistry, GOCART aerosol module, RRTMG radiation, planetary boundary layer (PBL) dynamics, fire plume-rise parameterisation (Paugam scheme), biomass burning emission modelling

**Spatial Data & GIS:** ArcGIS Pro, QGIS, NetCDF / GRIB / raster processing (Xarray, NCO), NASA FIRMS fire detection data, MODIS / VIIRS satellite burn scar analysis, ERA5 reanalysis, terrain / DEM analysis

**Programming & Statistical Analysis:** Python (NumPy, Pandas, Xarray, Matplotlib, SciPy, Scikit-learn), R (spatial analysis, statistical modelling), Bash scripting, process automation, Machine Learning for environmental systems

**Fire Science & Risk:** Fire behaviour analysis (wind-slope-fuel interactions), fire emission inventories (FINN, EDGAR-HTAP), aerosol optical depth & radiative forcing, air quality impact assessment (PM<sub>2.5</sub>, PM<sub>10</sub>, BC, OC, O<sub>3</sub>, NO<sub>x</sub>), fire risk quantification, fire weather indices

**HPC & Computing:** High-performance computing (HPC / Linux cluster), parallel WRF execution, large-scale simulation workflow management, Git / GitHub, technical documentation & scientific reporting

## Work Experience

**Civil Engineer, EEKI FOODS** March 2025 – May 2025

- Developed GIS-based **flood risk mitigation** strategies through hydraulic analysis, reducing hazard zones by **32%**.
- Produced grading plans, terrain models, and drainage designs using **Civil 3D** and spatial datasets.
- Conducted hydrological surveys supporting earthwork balancing, drainage, and environmental protection compliance.

**Engineering Intern, Jal Jeevan Mission, Ministry of Jal Shakti** May 2024 – July 2024

- Conducted GPS/QGIS field mapping across 40+ sites to support environmental assessment and reporting.
- Developed 3D hydraulic models for recharge structure evaluation, improving efficiency by **24.6%**.

**Research Intern, AICOE Project, Ministry of Education** May 2024 – July 2024

- Analysed and annotated 150+ hours of traffic data to build datasets for **AI-powered** infrastructure modelling.
- Developed automated validation pipelines and collaborated with interdisciplinary research teams.

## Research & Projects

**M.Tech Thesis: WRF-Chem-Fire Simulation of 2024 Uttarakhand Wildfire Episode** 2025 – 2026

*IIT Jodhpur* | *Supervisor: Dr. Amit Sharma* | *WRF-Chem-Fire, Python, NetCDF, ArcGIS, FIRMS, ERA5*

- Delivered the **first coupled WRF-Chem-Fire simulation** of the April 2024 Champawat-Purnagiri wildfire (Uttarakhand), covering fire-atmosphere-chemistry interactions at **1 km resolution** over the active fire zone and **3 km** for the broader Uttarakhand region.
- Implemented **two-way fire-atmosphere coupling** (Rothermel fire spread via level-set algorithm, fire mesh at **50 m effective resolution**), MOZCART photochemistry (chem\_opt=112), GOCART aerosol module, and RRTMG radiation with **aerosol radiative feedback** (ARF).
- Applied **Grid-FDDA nudging** (ERA5 reference,  $G = 3 \times 10^{-4} \text{ s}^{-1}$ ) — achieving wind speed NMB reductions of **29–38%** and IOA improvements of **40–44%** at validation stations versus free-running WRF.
- Quantified **fire emission impacts**: peak domain-averaged PM<sub>2.5</sub> rose to **71  $\mu\text{g m}^{-3}$**  (exceeding Indian NAAQS); BC enhanced  $\sim 9\times$ , OC  $\sim 10\times$ , NO<sub>2</sub>  $\sim 2.5\times$ ; NMB for PM<sub>2.5</sub> improved from **-19.2% to +6.4%** upon activating fire emissions.
- Isolated **aerosol radiative feedback (ARF)**: SWDOWN reduced by **15–25  $\text{W m}^{-2}$** , PBLH suppressed by **20–50 m**, PM<sub>2.5</sub> enhanced by **5–12  $\mu\text{g m}^{-3}$**  via reduced mixing volume; O<sub>3</sub> depleted by **1–3 ppbv** through UV photolysis attenuation.
- Characterised **diurnal fire spread dynamics**: two distinct peak spread periods (18 & 19 April), peak ROS  **$\sim 6.0 \text{ m s}^{-1}$** , areal growth  **$> 1.5 \text{ km}^2/\text{hr}$** ; total simulated burnt area  **$\sim 470 \text{ km}^2$**  in  $\sim 92$  hours — consistent with NASA FIRMS-derived burn scar of  **$\sim 414 \text{ km}^2$** .
- Processed multi-source spatial datasets: **ERA5** (meteorological IC/BC), **NASA FIRMS MODIS/VIIRS** (ignition points), **EDGAR-HTAP** anthropogenic emissions, **FINN v2.5** fire inventories, **CAM-Chem** chemical BCs, **MODIS MCD12Q1** land use — all in NetCDF/GRIB formats via Python workflows.
- Validated simulations against **CPCB CAQMS** real-time air quality observations (PM<sub>2.5</sub>, PM<sub>10</sub>, O<sub>3</sub>, NO<sub>2</sub>, temperature, wind) using NMB, nRMSE, Pearson  $r$ , and IOA metrics; performed domain resolution sensitivity analysis (1 km vs. 3 km).

- Conducted **terrain influence analysis** on fire spread: topographic wind channeling (Froude number analysis), upslope acceleration, valley flow deflection — explaining the complex multi-lobed fire perimeter geometry.

### Wildfire Risk Mapping & Fire Weather Analysis — Python, GIS

2024

*IIT Jodhpur | Python, ArcGIS, QGIS, NetCDF, Remote Sensing*

- Developed Python-based automated pipelines for ingesting and processing **ERA5 NetCDF** climate datasets (temperature, wind speed, relative humidity, precipitation) to derive fire weather indices and identify extreme fire weather periods.
- Performed **spatial fire risk analysis** by integrating MODIS land cover, digital elevation models, and meteorological rasters in ArcGIS/QGIS to map fuel type distribution and **fire hazard exposure zones**.
- Applied **statistical trend analysis** and time-series visualisation on historical fire season data to identify drivers of pre-monsoon wildfire intensification over the Indian Himalayan Region.
- Generated **automated validation and reporting scripts** in Python comparing model outputs against satellite-derived fire observations (NASA FIRMS) and CPCB ground station data.

### Urban Flood & Drainage Modelling — HEC-RAS, GIS

2024

*IIT Jodhpur | HEC-RAS, AutoCAD, Civil 3D, ArcGIS*

- Designed urban drainage and stormwater conveyance systems using HEC-RAS and GIS spatial data, optimising drainage capacity by **27%** and reducing earthwork by **19%**.
- Produced construction-ready grading and drainage drawings; conducted hydrological field surveys.

### Building Energy Efficiency & Fire Safety Audit

2023

- Estimated HVAC thermal loads by calculating HDD/CDD and evaluating heat gains for energy efficiency.
- Conducted a fire safety compliance audit across G+3 academic and hostel facilities against NBC 2016 standards.

### Positions of Responsibility & Achievements

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- **Department Rank 1:** B.Tech–M.Tech Dual Degree, Civil & Infrastructure Engineering, IIT Jodhpur
  - **M.Tech CGPA 9.5 / 10** — M.Tech thesis constitutes the first published WRF-Chem-Fire coupled simulation for the Champawat–Purnagiri region, Uttarakhand
  - **Festival Chief, Ganeshotsav:** Led 200+ volunteers in cultural operations and campus logistics
  - **Head, Public Relations, EDIFICIO** (IIT Jodhpur Civil Engineering Festival)